Roll Number:______________________

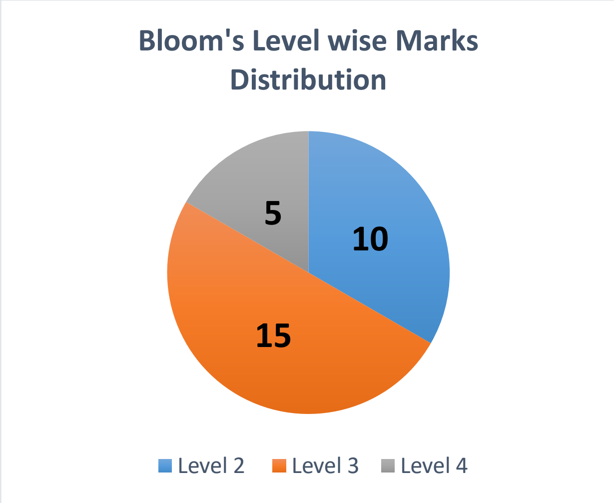
Thapar Institute of Engineering &Technology, Patiala, Punjab, India

Department of Electronics and Communication Engineering

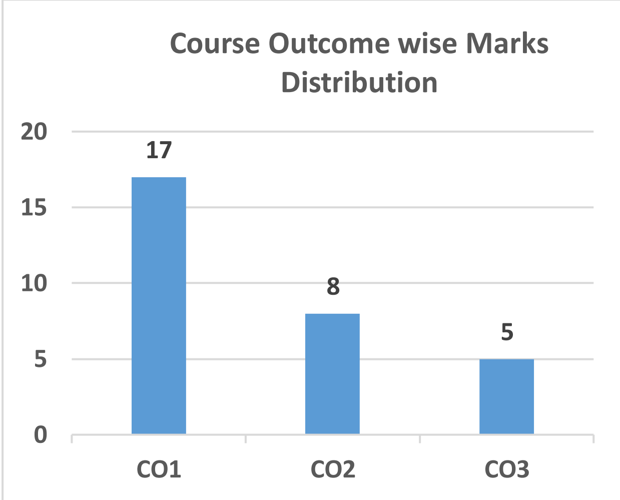
M. Tech. (VLSI Design) Semester:- II	Course Code:- PVL207 Course Name:- Low Power System Design
March 20th, 2024; **MST**	Wednesday, 03:00 P.M. – 05:00 P.M.
Time: 02 hrs., Max. Marks: 30	**Name of Faculty: Dr. Mohammad Faseehuddin**

Note: Attempt all questions (Write mathematical expressions to support your answers wherever applicable)

Q.No.	Questions	Marks	CO	BL
Q.1	Discuss various sources of power dissipation in CMOS digital circuits. Also, discuss the different types of power dissipation that takes place in a digital IC.	**(5)**	**CO1**	**L2**
Q.2(a)	Explain the Sub-threshold leakage in a MOS transistor with proper mathematical expression. How it can be minimized?	**(3)**	**CO2**	**L3**
Q.2(b)	Discuss hot electron effect and velocity saturation. What implications do they have on the circuit performance?	**(2)**	**CO1**	**L2**
Q.3(a)	Discuss the constant voltage scaling technique. What effect does it have on (i) power dissipation (ii) power density (iii) delay (iv) area (v) W, L, V_{DD} (supply voltage).	**(3)**	**CO2**	**L2**
Q.3(b)	Differentiate between dynamic and static logic circuits. What advantages do dynamic circuits have over static logic circuits?	**(2)**	**CO2**	**L3**
Q.4	Discuss different techniques to reduce power dissipation of digital circuits in brief.	**(5)**	**CO1**	**L3**
Q.5	Explain Variable threshold voltage CMOS (VTCMOS) approach and multi-threshold-voltage CMOS (MTCMOS) in detail with a design example.	**(5)**	**CO3**	**L4**
Q.6 (a)	Design D latch using CMOS technology and explain its operation.	**(3)**	**CO1**	**L3**
Q.6 (b)	Design a SR Latch using NOR implementation in CMOS technology.	**(2)**	**CO1**	**L3**



Bloom's Level wise Marks Distribution
5
10
15
Level 2
Level 3
Level 4



Course Outcome wise Marks Distribution
20
15
10
5
0
17
8
5
CO1
CO2
CO3